

PRACTICAL TEST AUTOMATION WITH PLAYWRIGHT

Debugging & Trace Viewer Guide

Diagnose failures fast with Playwright's debugging tools.

Introduction

When a test fails, good debugging tools turn confusion into a quick fix. Playwright has excellent built-in tooling. This guide covers the tools you will use daily.

Your Debugging Toolkit

Tool	Command / use
UI mode	<code>npx playwright test --ui</code> — watch, step, time-travel
Headed mode	<code>--headed</code> — see the browser run
Debug mode	<code>--debug</code> — step through with Inspector
Trace viewer	<code>npx playwright show-trace trace.zip</code>
Codegen	<code>npx playwright codegen <url></code> — record + get locators

Reading a Trace

- Timeline of every action with before/after DOM snapshots
- Network requests and console logs
- The exact point and reason a step failed

Worked Example

A test fails in CI. You enable trace: 'on-first-retry', re-run, download the trace artefact, open it with show-trace, and see the click happened before the element was ready — pointing you straight to the fix.

Practical Exercise

Deliberately break a locator, run with traces enabled, open the trace and identify exactly where and why it failed.

Best Practices

- Enable traces on first retry in CI.
- Use UI mode while developing tests.
- Use codegen to find robust locators quickly.

Common Mistakes

- Guessing at failures instead of opening the trace.
- No artefacts captured in CI.
- Re-running blindly hoping it passes.

INSIDE STLC PRO TIP

The trace viewer is the single best debugging feature in Playwright. Before changing any code to fix a failure, open the trace — it usually shows you the cause in seconds.